Technical Data Sheet

NebuLink Module
NLN0721 – Wi-SUN Module

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ds_nln0721_3.0

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Preface

Privacy Information

Table below shows the revision history of Wi-SUN module NLN0721 data sheet.

Revision History

Version	Date	Description of Changes
1.0	September, 2023	Internal Release
2.0	November, 2023	Initial Release
3.0	May, 2024	Regulatory Approval details added.

Typographic Conventions

This data sheet uses the typographic conventions as shown below:

Visual Cue	Meaning
Bold Type with Initial Capital letters	All headings and Sub headings Titles in a document are displayed in bold type with initial capital letters; Example: General Description, Features.
Bold Type with Italic Letters	All Definitions, Figure and Table Headings are displayed in Italics. Examples: ***Figure 1-1. NebuLink Node NLN0721, Table 1-1. NebuLink Node NLN0721 Specifications.***
1., 2.	Numbered steps are used in a list of items, when the sequence of items is important. such as steps listed in procedure.
•	Bullets are used in a list of items when the sequence of items is not important.

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1 Introduction

The NLN0721 is an ideal sub-GHz solution for "Internet of Things" applications such as smart metering, smart lighting, building automation, and smart homes. This sub-GHz module based on EFR32FG28 offers long-range and mesh connectivity for multi-hop connections. This single-die, multi-core solution provides industry-leading security features, low power consumption with fast wake-up times, and integrated power amplifiers to enable the next level of secure connectivity for IoT devices. A large memory footprint enables flexible solutions for dynamic protocol support. The compact size of the module makes it an ideal choice for any IoT device. With a lifespan and support of 10 years, this module is the most appropriate choice for your IoT end devices.

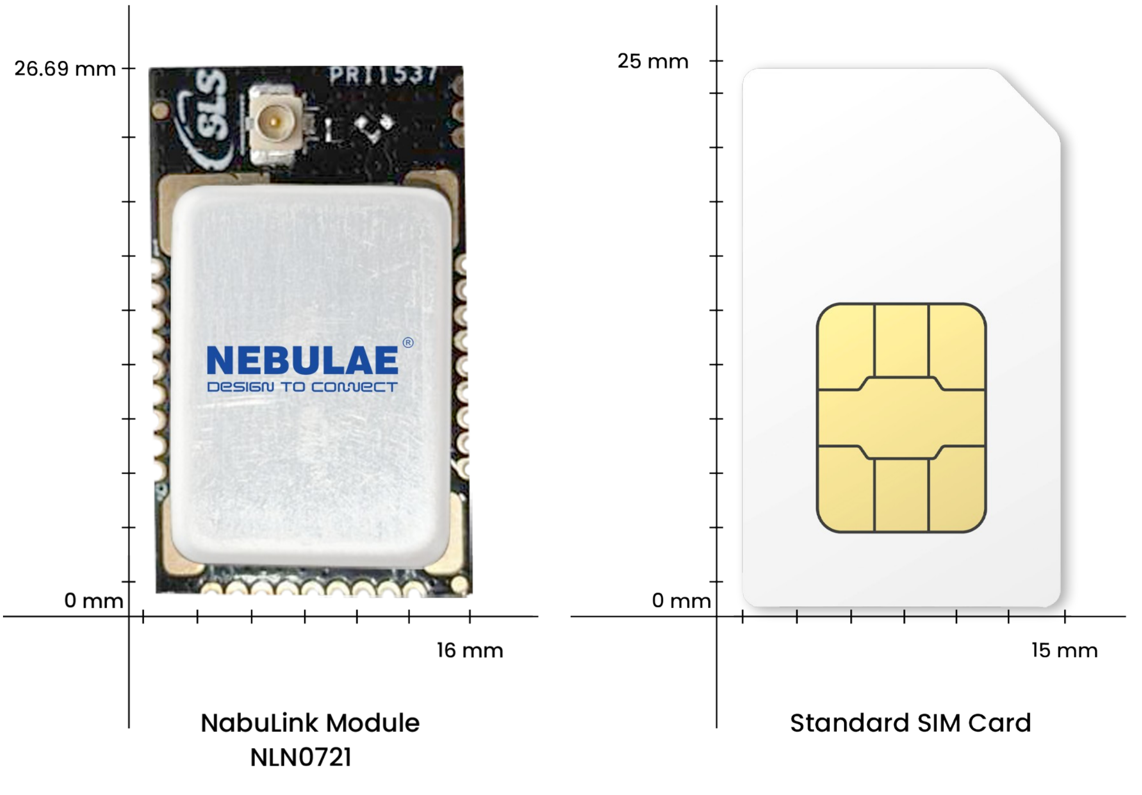

Figure 1: NLN0721 – Wi-SUN based NebuLink Module

2 Key Features

- Frequency Band: Supports EU 863-870, US 902-928, IN865
- TX power: Up to +20 dBm
- Protocol Support: Amazon Side-walk, WM-BUS, Wi-SUN
- Power Supply: 1.8V to 3.6V
- Operating Temperature Range: -20°C to +75°C
- Flash Program Memory: Up to 1024 kB
- RAM Data Memory: Up to 256 kB
- Hardware Cryptographic Acceleration: AES128/192/256
- OTA (over-the-air): Firmware update over-the-air (FUOTA)
- Energy-efficient radio core with low active and sleep currents
- Sensitivity: -108 dBm @100kbps on 865-867 MHz, Wi-SUN FSK
- Serial-wire debugging (SWD)
- I/Os: Up to 21 I/Os on the Edge Plated holes of the module High-performance 32-bit 78 MHz ARM Cortex®-M33 with DSP instruction and floating-point unit for efficient signal processing
- The module can be directly soldered to any device board.
- RF shield on the module to protect it from external electromagnetic radiation and interference

3 Brief Specification

Table – 1 Module Size

Module	Chip	Power (dBm)	Antenna	Dimension (mm) LxWxH
NLN0721	EFR32FG28	+20	u.FL/Feed line	26.69 X 16 X 3.65

4 Application

The applications include, but are not limited to:

- Smart metering
- Smart street lighting
- Smart waste management
- Smart City Network
- Smart City Traffic Control
- Smart agriculture
- Building automation & monitoring
- Lighting controls
- Inventory management
- Environmental monitoring
- Industrial monitoring
- Machinery condition and performance monitoring

5 Ordering Information

Table – 2 Ordering Information	
Part Number	**RF Output**
PL0NL0721I100	u.FL
PL0NL0721S100	Feed line

6 Block Diagram

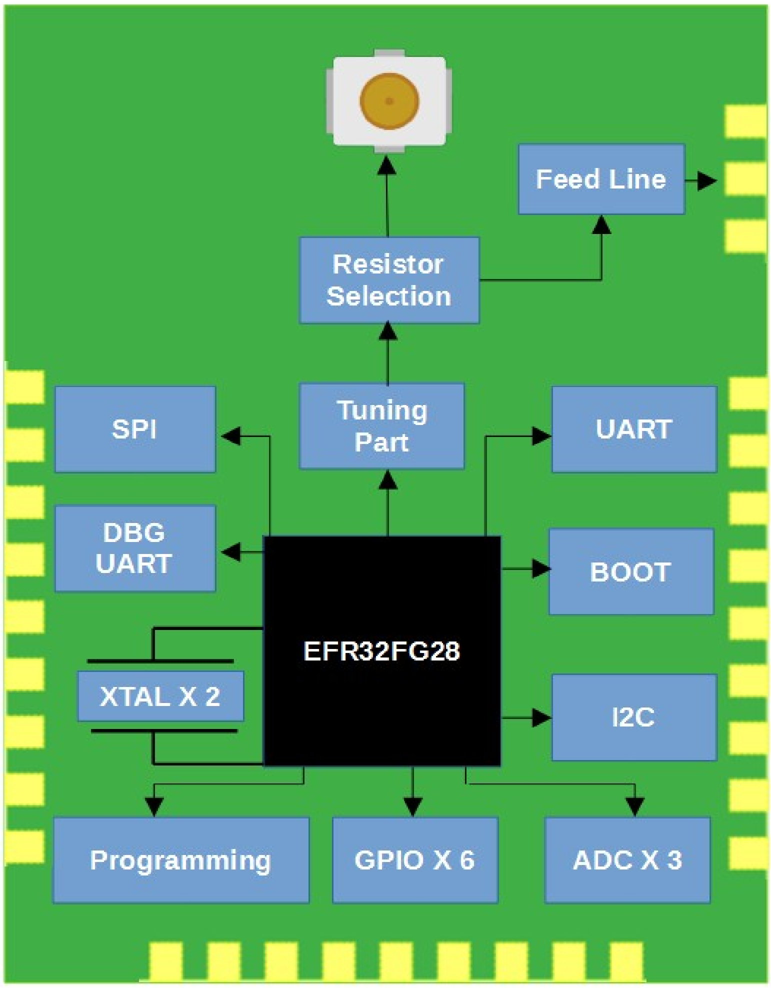

Figure 2: Block Diagram

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7 Pin Details

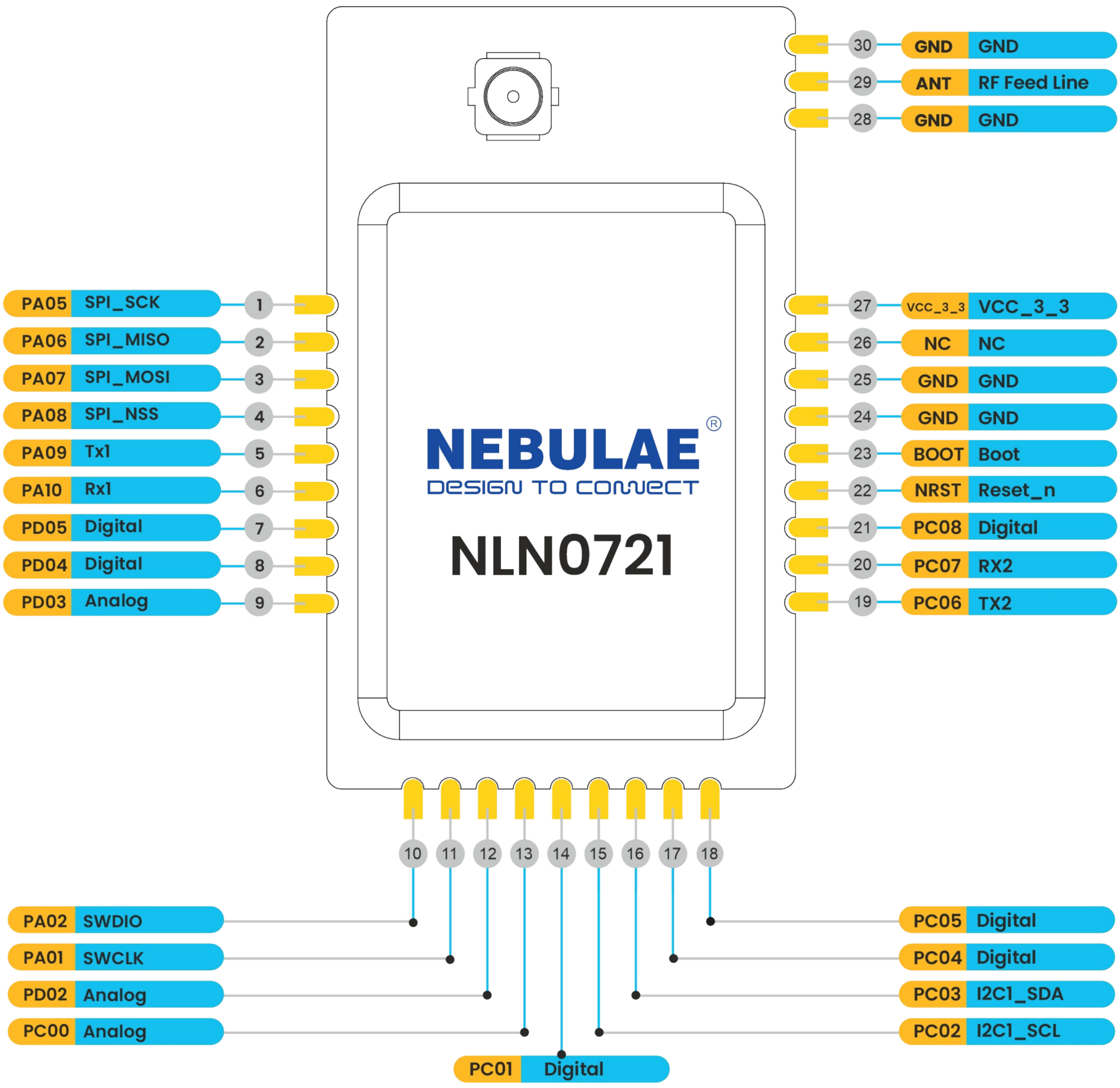

Figure 3: Pinout Diagram

Table – 3 Pin Details		
Pin Number	**Require Pins for SLS**	**Description**
1	PA05	SPI_SCK
2	PA06	SPI_MISO
3	PA07	SPI_MOSI
4	PA08	SPI_NSS
5	PA09	TX1
6	PA10	RX1
7	PD05	Digital
8	PD04	Digital
9	PD03	Analog
10	PA02	SWDIO
11	PA01	SWCLK
12	PD02	Analog
13	PC00	Analog
14	PC01	Digital
15	PC02	I2C1_SCL
16	PC03	I2C1_SDA
17	PC04	Digital
18	PC05	Digital
19	PC06	TX2
20	PC07	RX2
21	PC08	Digital
22	NRST	Reset_n
23	BOOT	Boot
24	GND	GND
25	GND	GND
26	NC	NC
27	VCC_3_3	VCC_3_3
28	GND	GND
29	ANT	RF Feed line
30	GND	GND

7.1 Peripherals

Table – 4 Peripherals	
Interface	**Quantity**
I2C	1
SPI	1
U(S)ART	1
LPUART	1

Table below provides the information about the usage of the I/Os available in NLN0721.

Table – 5 I/O Details	
Usage	**Quantity**
Digital Pin	Up to 21 pins
ADC	3 pins
Comparators	2 pins
DAC	1 pin

8 Operating Characteristics

Table – 6 Operating Characteristics

Parameter	Minimum	Typical	Maximum	Unit
Temperature	–20		70	°C
Supply Voltage (VDD)	1.8	3.3	3.8	V
Supply Voltage Ripple or Noise (VDD)			10	mV
Frequency IN	865		868	MHz
TCXO Frequency		39		MHz
XTAL Frequency		32.768		KHz
XTAL Frequency Tolerance		±20ppm		
Power consumption(standby mode)		6.95		mA

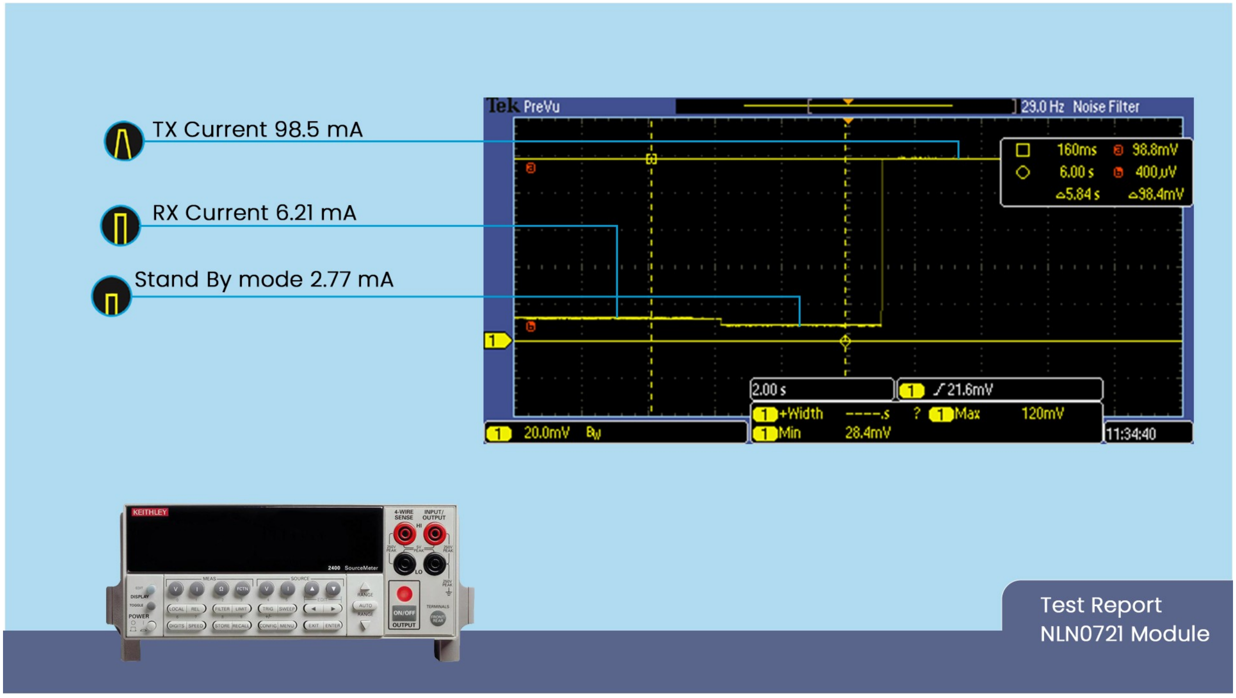

Figure 4: Operating Characteristic Test Report

9 RF Characteristic

Table – 8 RF Characteristic				
Parameter	**Minimum**	**Typical**	**Maximum**	**Unit**
Output RF level (High PA)		+19.5	20	dBm
Power consumption, TX Mode	-	-	98.5	mA
Power consumption, RX Mode			6.21	mA
Sensitivity (865Mhz, 100kbps, No FEC)		-108		dBm
Sensitivity (865Mhz, 50kbps, No FEC)		-109.5		dBm
Sensitivity (863-870MHz, 50kbps,no FEC)		-109.5		dBm
Frequency Band	IN865			
Modulation	Wi-SUN FSK/ 2GFSK			

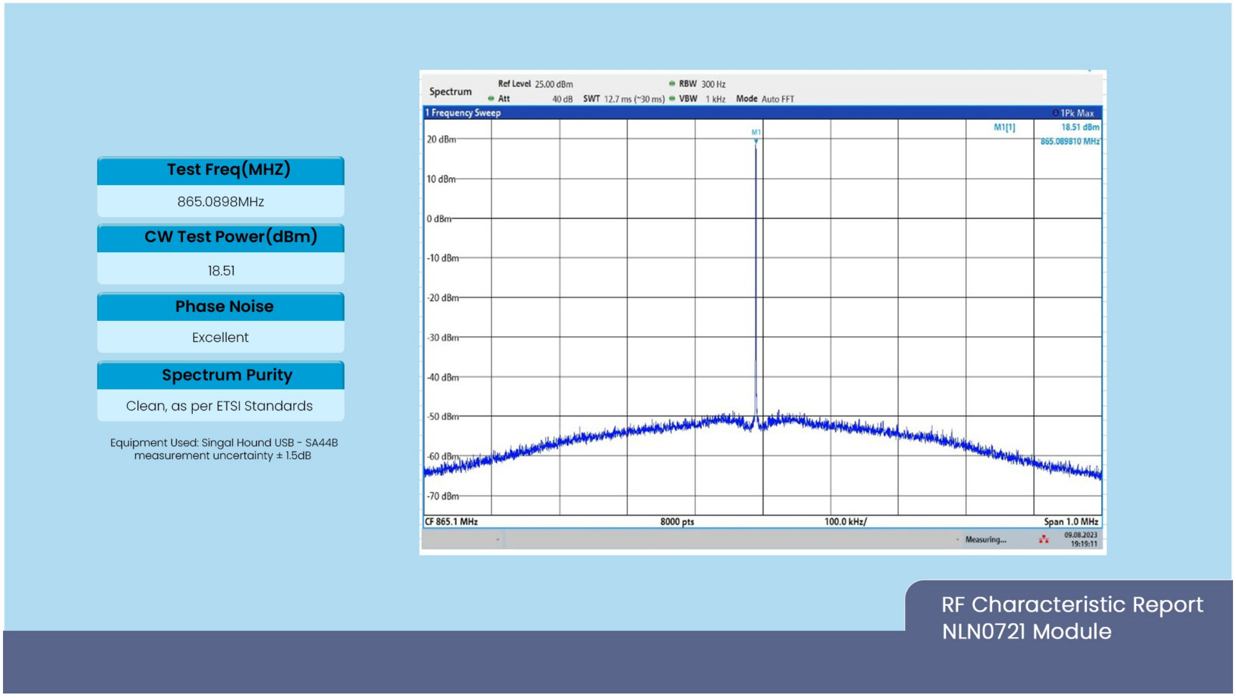

Figure 5: RF Characteristic Report

10 Programming of NLN0721 (EFR32FG28)

1. To generate the code for NLN0721(EFR32FG28) download the simplicity-studio v5 tool from below link:

https://www.silabs.com/developers/simplicity-studio

2. After downloading Update simplicity-studio v5.

3. Download Gecko SDK Suite for Wi-SUN using simplicity-studio v5.

4. To program the binary, you need to install Simplicity Commander using Simplicity Studio or by downloading one of the following standalone versions and then completing the installation:

https://www.silabs.com/developers/simplicity-studio#commander

OR

https://www.silabs.com/documents/public/software/SimplicityCommander-Linux.zip

https://www.silabs.com/documents/public/software/SimplicityCommander-Mac.zip

https://www.silabs.com/documents/public/software/SimplicityCommander-Windows.zip

5. Use below pins to program the NLN0721(EFR32FG28) module:

1. GND (Pin 23, 24)
2. 3V3 (Pin 27)
3. SWDIO (PA02) (Pin 10)
4. SWCLK (PA01) (Pin 11)
5. NRST (Pin 12)

6. Wireless Starter Kit (WSTK) can be used to program NLN0721(EFR32FG28) module. You need to set WSTK in OUT Mode. Your custom board should be detected. Once detected, flash/program it.

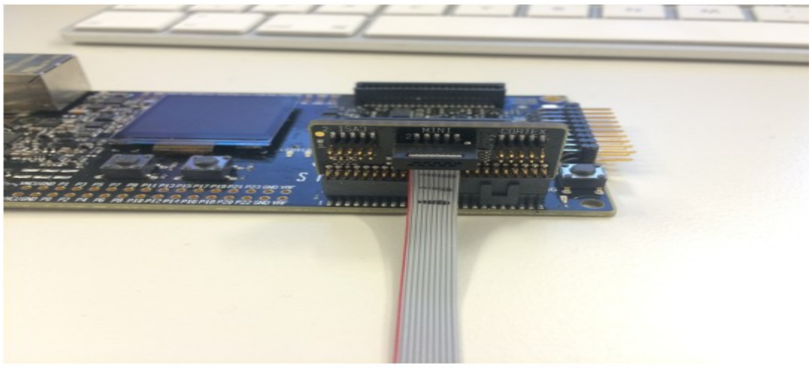

Figure 6: Simplicity Debug Adaptor Board

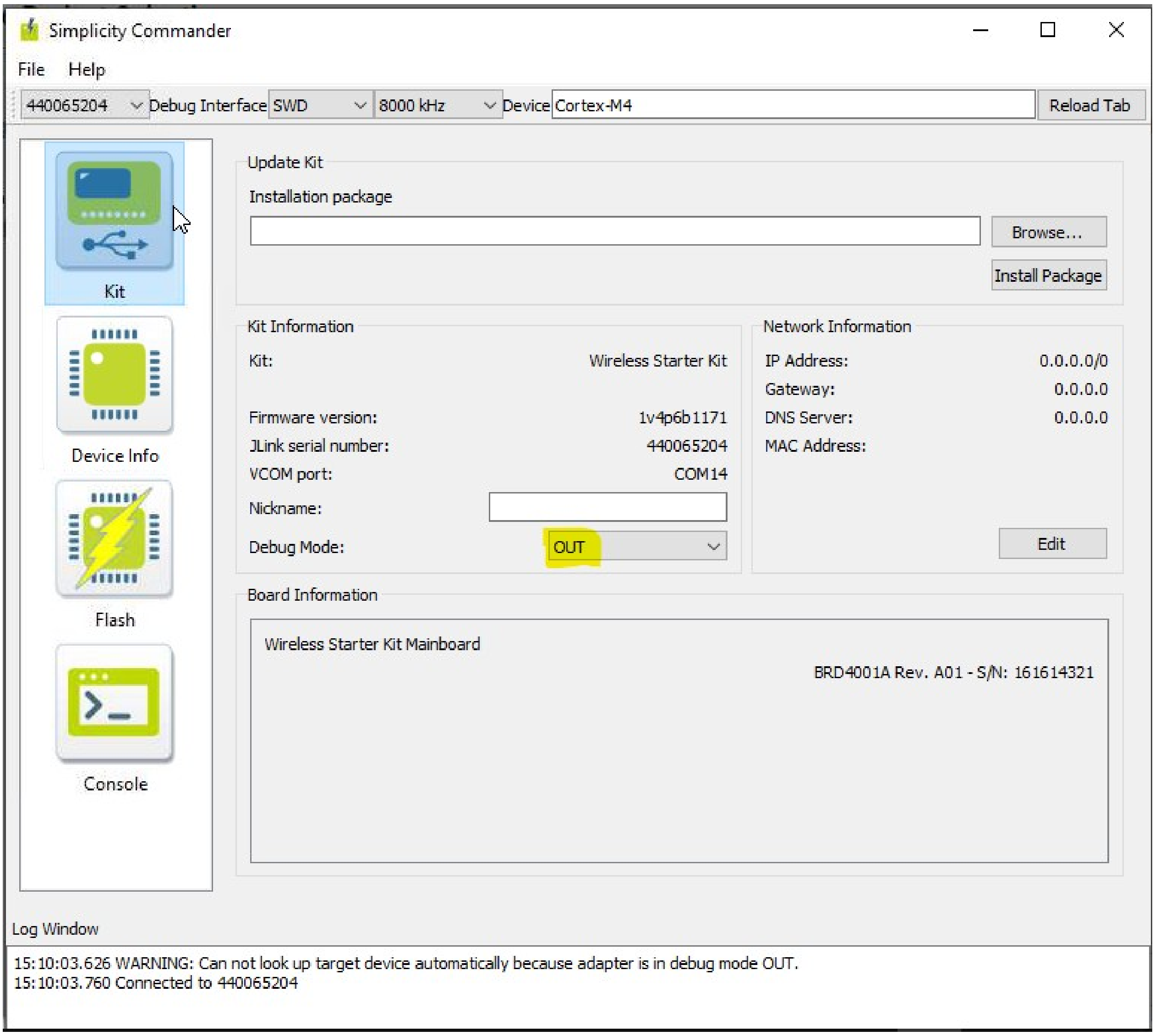

Figure 7: Simplicity Commander Screen

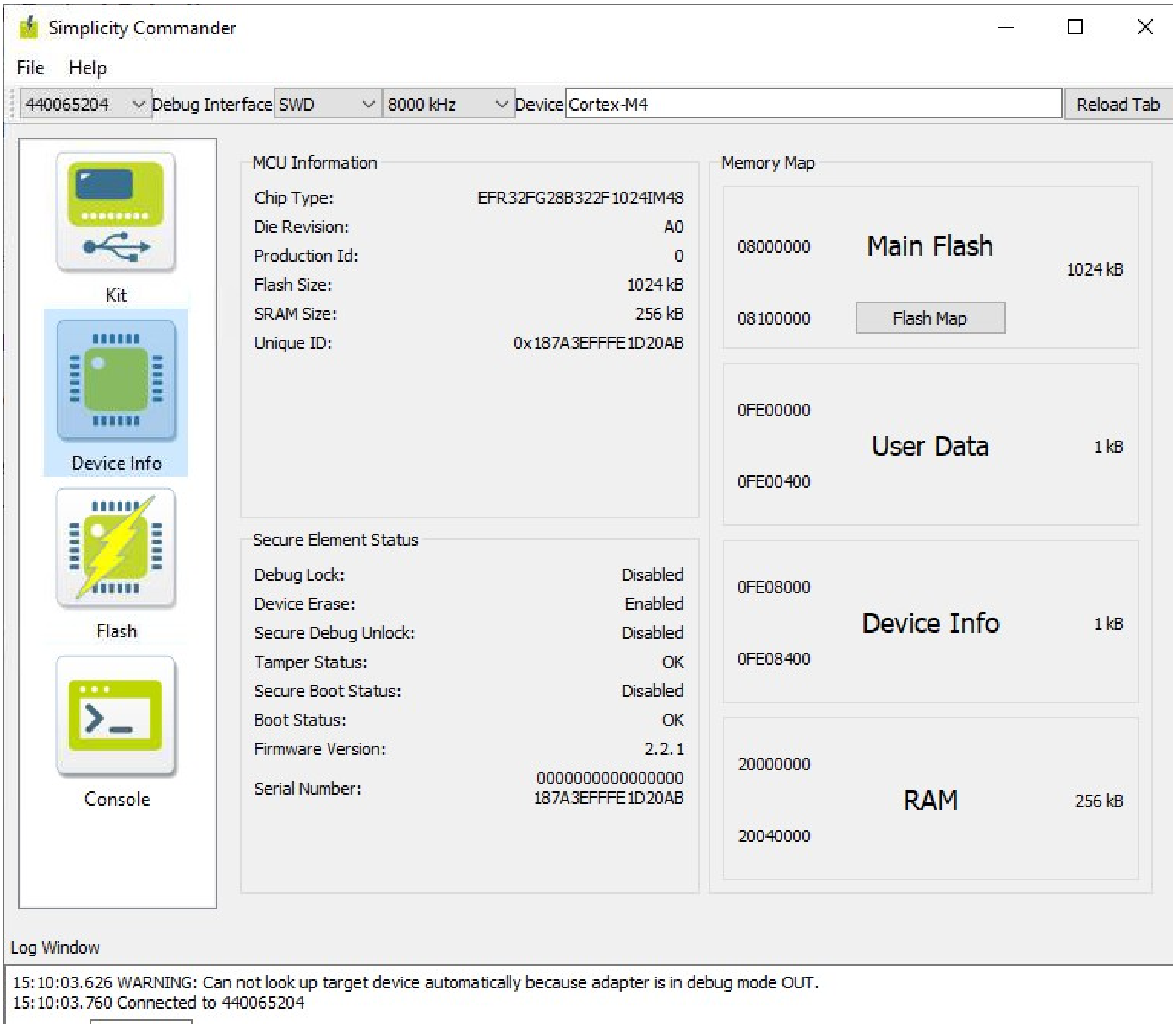

Figure 8: Simplicity Commander Screen

11 Reference Schematic

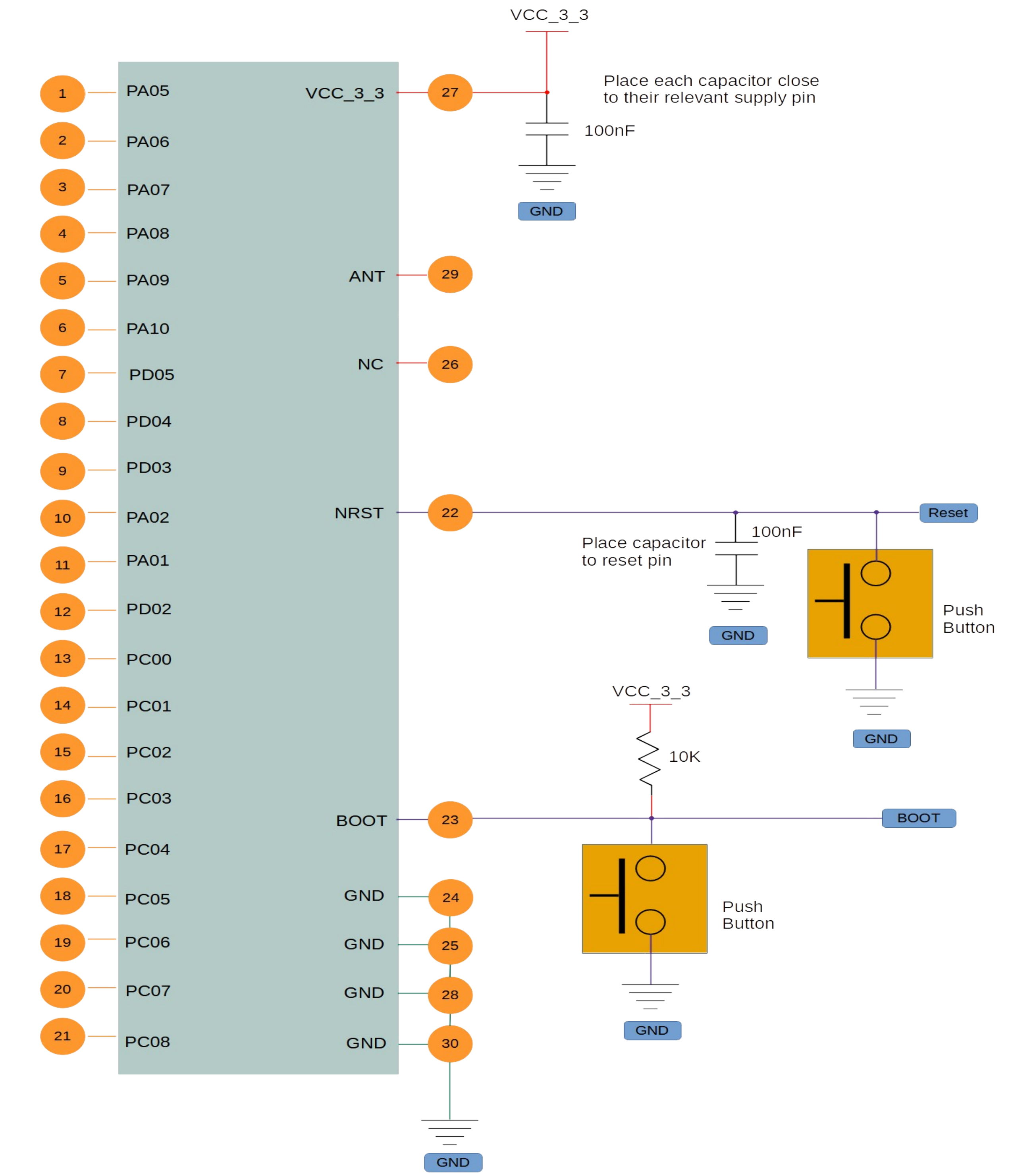

Figure 9: Reference Schematic

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12 Package Outline

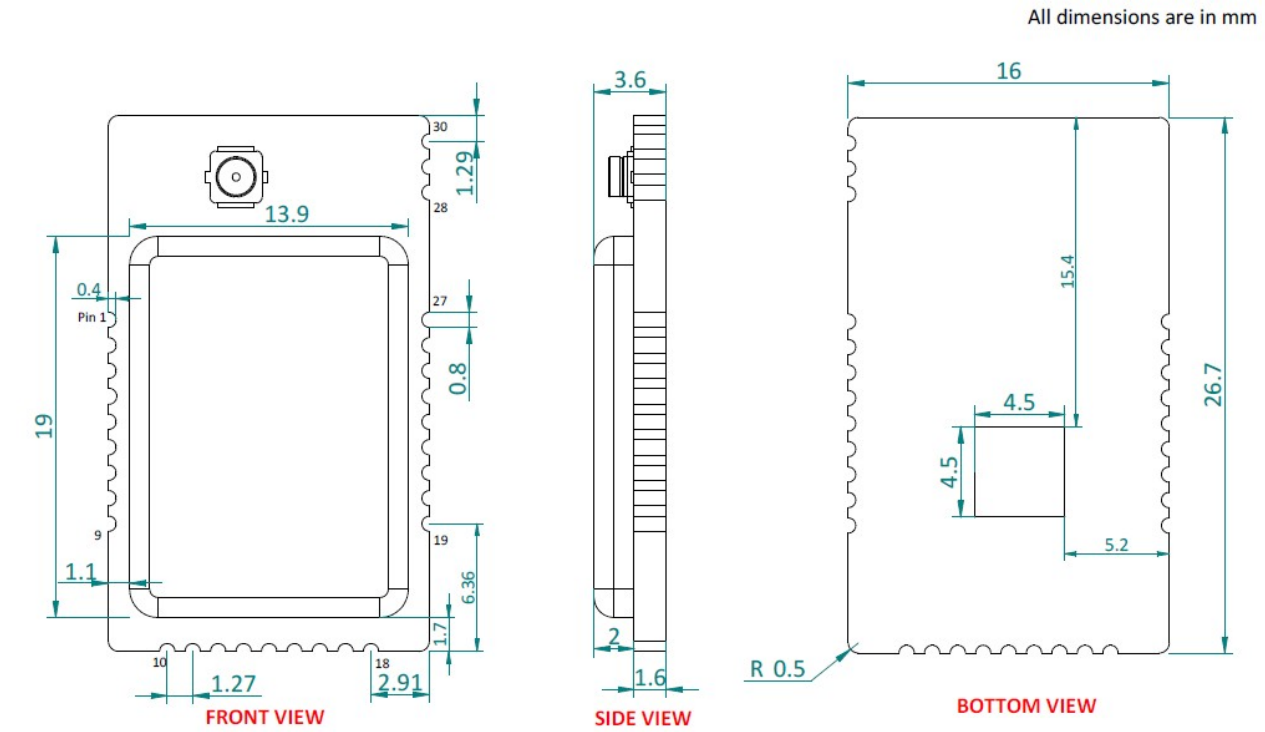

Figure 10: Size & Dimensions

13 Recommanded Footprint

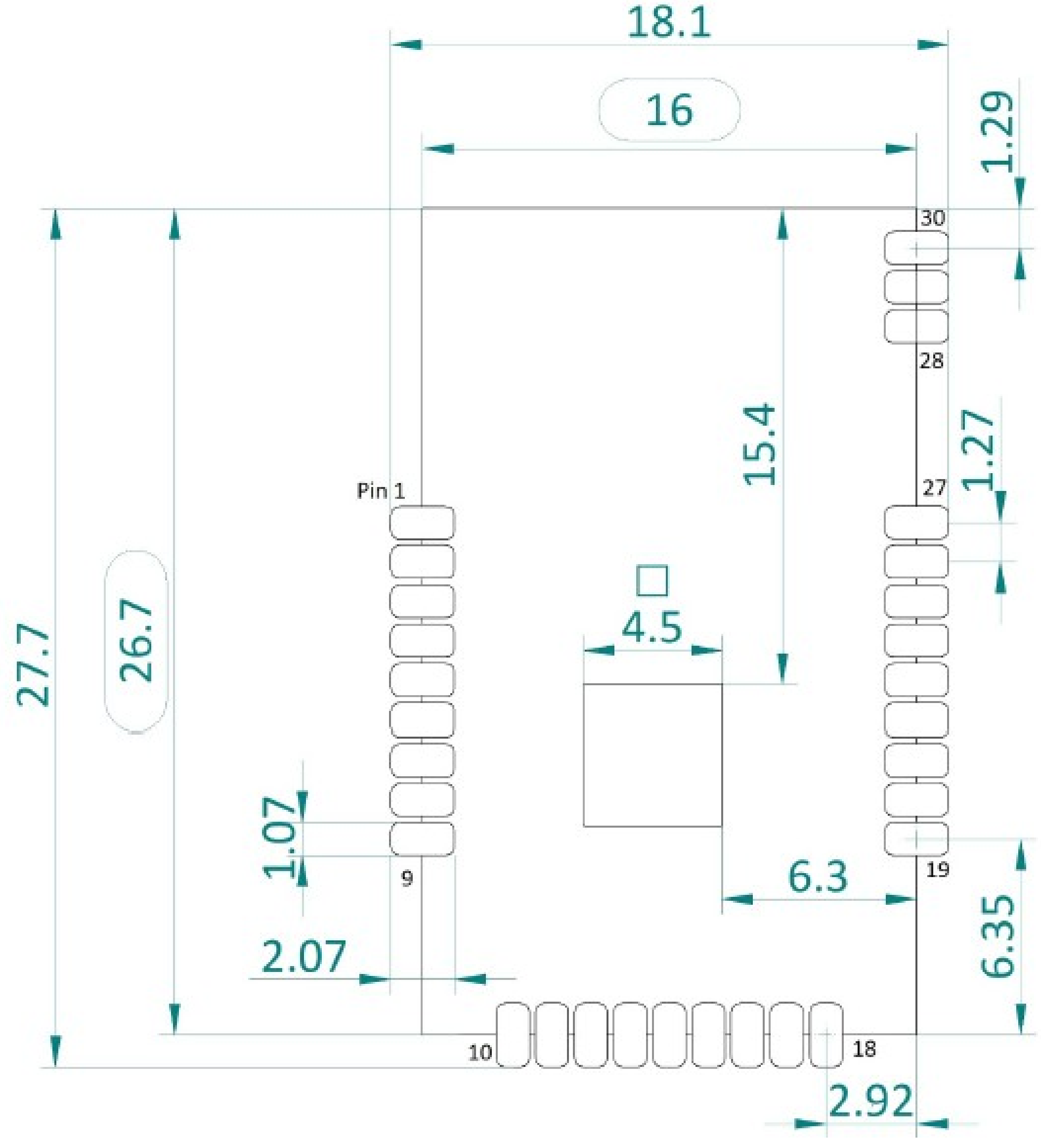

Figure 11: Recommanded Footprint

14 Layout Guidelines

The PCB design has used enough ground area which increases the antenna efficiency. This allows stand-alone operation without any additional ground plane however care must be taken when mounting this module onto another PCB.

"The most critical aspects of radio frequency (RF) circuitry are addressed and it is highly recommended to follow these design guidelines to achieve best RF performance."

1) Keep digital and power traces/wiring away from RF circuits; clock and high-frequency circuits are the principal sources of interference and radiation, and must be placed separately and away from sensitive circuits.

2) The RF line has a 50-ohm impedance, so the line width should be as wide as feasible to match the impedance requirements.

3) Avoid parallel routing and crossing of RF traces and from under signal pads.

4) Provide good solid ground by using multiple vias.

5) The recommended practice is to use a solid (continuous) ground plane on Layer 2, assuming Layer 1 is used for the RF components and transmission lines.

6) A common practice is to use a "star" configuration for the power-supply routes.

7) Most ICs require a solid ground plane on the component layer (top or bottom of PCB) directly underneath the component. This ground plane will carry DC and RF return currents through the PCB to the assigned ground plane.

15 Placement Guidelines

The PCB design has used enough ground area which increases the antenna efficiency. This allows stand-alone operation without any additional ground plane however care must be taken when mounting this module onto another PCB.

The area around the antenna must be kept clear of any conductive or metal parts for an absolute minimum of 20 mm. Any conductive objects close to the antenna could severely disrupt the antenna pattern resulting in deep nulls and high directivity in some directions. This is true for all layers of the PCB and not just the top layer.

It is recommended that the module is mounted on the edge of the host PCB with the antenna exposed to free space. It is permitted for PCB material to be below the antenna structure of the module as long as no copper traces or planes are on the host PCB in that area.

Never place any component, planes, mounting screws, or traces in the antenna keep-out area across all layers. The actual keep-out area depends on the antenna used

Do not place the antenna close to the plastic in the industrial design. Plastic has a higher dielectric constant than air. Proximity of the plastic to the antenna results in the antenna's seeing a higher effective dielectric constant. This increases the electrical length of the antenna trace and reduces the resonant frequency.

The battery cable or any electrical interface cable must not cross the antenna trace or Antenna.

The antenna must not be covered by a metallic enclosure completely. If the product has a metallic casing or a shield, the casing must not cover the antenna. No metal is allowed in the antenna near-field.

The orientation of the antenna should be in line with the final product orientation so that the radiation is maximized in the desired direction

There must not be any ground directly below the antenna.

16 Soldering Guidelines

For re-flow soldering, it is recommended to follow the reflow profile in below figure as a guide, as well as the paste manufacturer's guidelines on peak flow temperature, soak times, time above liquid and ramp rates.

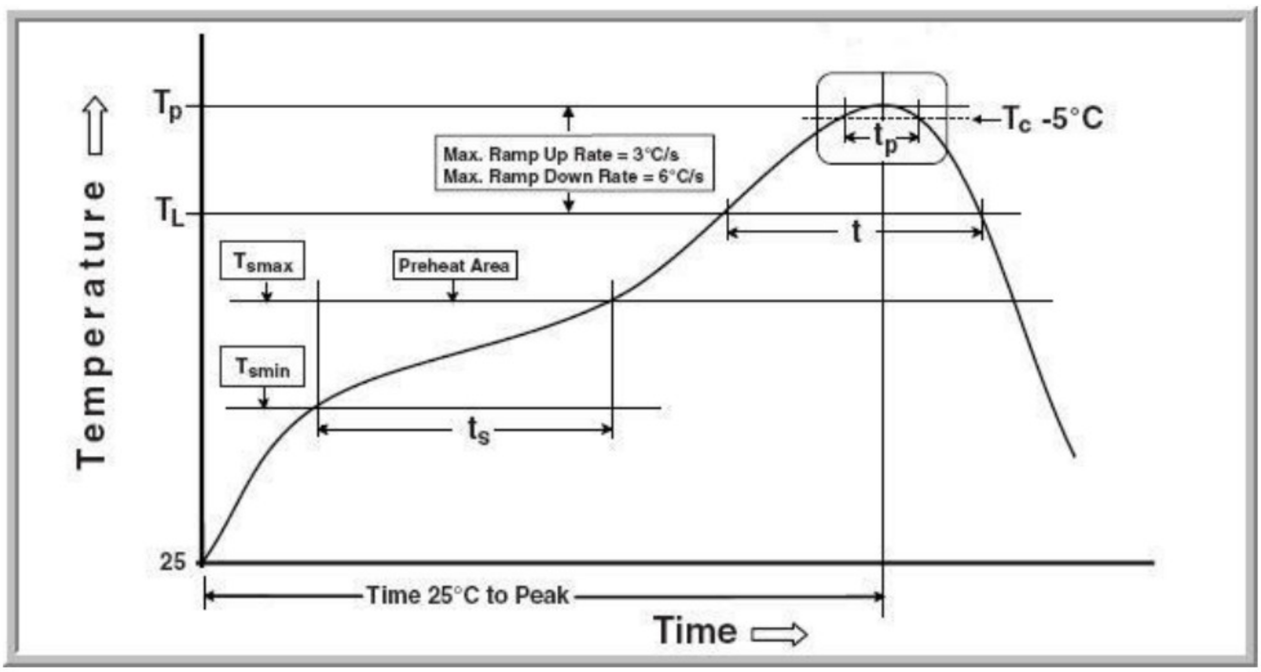

Figure 12: Reflow

Table – 10 Soldering Information		
Profile Feature		**Pb-Free Assembly**
Average Ramp-Up Rate (TSmax to TP)		3°C / second max.
Preheat	Temperature Min (Tsmin) Temperature Max (Tsmax) Time (tSmin to tSmax)	150°C 200°C 60-180 seconds
Time maintained above	Temperature (TL) Time (tL)	217°C 60-150 seconds
Peak/Classification Temperature (TP)		245°C
Time within 5°C of the specified Peak Temperature (tP)		20-40 seconds
Ramp-Down Rate (TP to TL)		6°C / second max.
Time 25°C to Peak Temperature		8 minutes max.

17 Regulatory Approval

The module, NLN0721, is WPC-certified.